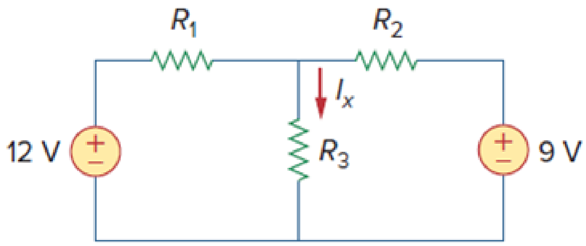
Chapter 3, Problem 1P

Problem

Using Fig. 3.50, design a problem to help other students better understand nodal analysis.

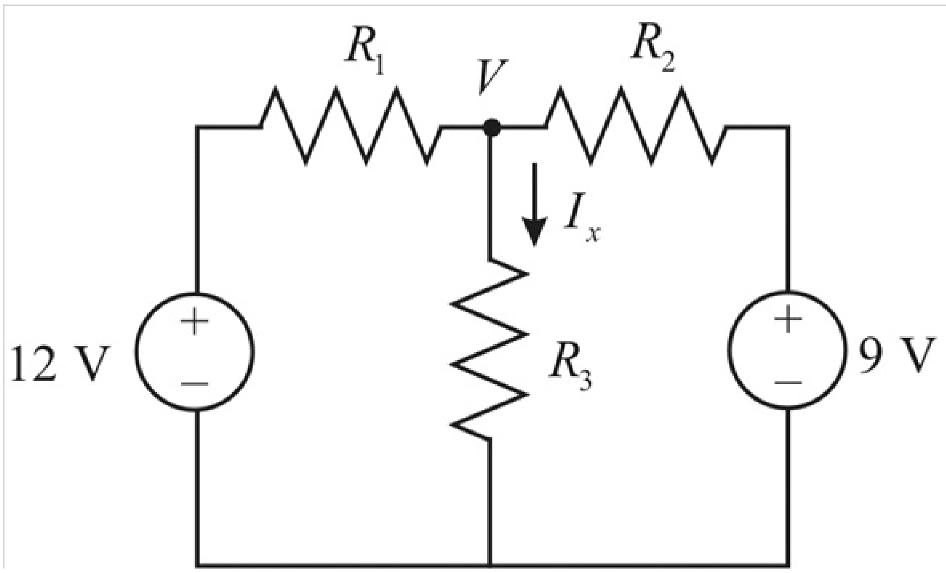
Figure 3.50:



Step-by-step solution

Step 1 of 3

Given circuit is



Step 2 of 3

By nodal analysis's we have the equation

$$\frac{V-12}{R_1}+\frac{V}{R_3}+\frac{V-9}{R_2}=0 \quad \ldots\ldots (1)$$

From the circuit we have $I_x=\frac{V}{R_3}$ ...... (2)

Solving the equation (1)

$$\frac{R_2R_3(V-12)+R_1R_2V+R_1R_3(V-9)}{R_1R_2R_3}=0$$

$$R_2R_3V-12R_2R_3+R_1R_2V+R_1R_3V-9R_1R_3=0$$

$$V(R_2R_3+R_1R_2+R_1R_3)=9R_1R_3+12R_2R_3$$

Step 3 of 3

$$V = \frac{9R_1R_3 + 12R_2R_3}{R_2R_3 + R_1R_2 + R_1R_3}$$

$$V = \frac{3R_3(3R_1 + 4R_2)}{R_2R_3 + R_1R_2 + R_1R_3}$$

From (2) we have $I_x = \frac{V}{R_3}$

Then $\boxed{I_x = \frac{3(3R_1R_3 + 4R_2R_3)}{R_3(R_2R_3 + R_1R_2 + R_1R_3)}}$